

Problem

Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$. If $v_1 = 10 \cos(50t - \pi/3)$ V and $v_2 = 12 \cos(50t + 30^\circ)$ V, find v .

Step-by-step solution

Step 1 of 3

Consider the following voltage, v_1 :

$$v_1 = 10 \cos\left(50t - \frac{\pi}{3}\right) \text{ V}$$

Here,

The amplitude, $V_m = 10$ V,

The angular frequency, $\omega = 50$ rad/s,

The phase angle, $\phi = -\frac{\pi}{3}$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the voltage, v_1 .

$$v_1(t) = 10 \angle -\frac{\pi}{3} \text{ V}$$

The voltage, v_1 is in polar form.

Use polar to rectangular transformation for voltage, v_1 .

$$\begin{aligned} v_1(t) &= 10 \angle -\frac{\pi}{3} \\ &= 10 \left[\cos\left(-\frac{\pi}{3}\right) + j \sin\left(-\frac{\pi}{3}\right) \right] \\ &= 10 [0.5 - j0.866] \\ &= (5 - j8.66) \text{ V} \end{aligned}$$

Step 2 of 3

Consider the following voltage, v_2 :

$$v_2 = 12 \cos(50t + 30^\circ) \text{ V}$$

Here,

The amplitude, $V_m = 12 \text{ V}$,

The angular frequency, $\omega = 50 \text{ rad/s}$,

The phase angle, $\phi = 30^\circ$.

The phasor is a complex number that represents the amplitude and phase of a sinusoid.

The following is the phasor for the voltage, v_2 .

$$v_2(t) = 12 \angle 30^\circ \text{ V}$$

The voltage, v_2 is in polar form.

Use polar to rectangular transformation for voltage, v_2 .

$$\begin{aligned} v_2(t) &= 12 \angle 30^\circ \\ &= 12 [\cos(30^\circ) + j \sin(30^\circ)] \\ &= 12 [0.866 + j0.5] \\ &= (10.39 - j6) \text{ V} \end{aligned}$$

Step 3 of 3

Consider that two voltages, v_1 and v_2 appear in series, so that the following is the sum of these voltages.

$$v = v_1 + v_2$$

Substitute the values of voltages, v_1 and v_2 to calculate the value of v .

$$\begin{aligned} v &= (5 - j8.66) + (10.39 + j6) \\ &= (5 + 10.39) - j(8.66 - 6) \\ &= (15.39 - j2.66) \text{ V} \end{aligned}$$

Represent the voltage, v in phasor form.

$$\begin{aligned} v &= (15.39 - j2.66) \text{ V} \\ &= \sqrt{15.39^2 + 2.66^2} \angle \tan^{-1} \left(\frac{-2.66}{15.39} \right) \\ &= \sqrt{244} \angle \tan^{-1} (-0.173) \\ &= 15.62 \angle -9.8^\circ \text{ V} \end{aligned}$$

The following is the general expression for the voltage, v .

$$v = 15.62 \cos(50t - 9.8^\circ) \text{ V}$$

Thus, the voltage, v is $\boxed{15.62 \cos(50t - 9.8^\circ) \text{ V}}$.